

PERIMETER AND AREA

We learnt the concept of perimeter and area in previous class. Let us recall.

Perimeter is the length of boundary of a plane closed figure and area is the measurement of surface of plane closed figure.

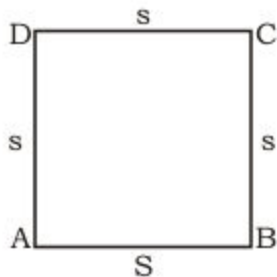
Find perimeter and area of a square and a rectangle**Perimeter of square:**

We know that the sum of measures of all the sides of a square is its perimeter.

In the figure, the length of each side of square is S units.

Perimeter = Sum of all sides

$$\begin{aligned}\text{So, Perimeter} &= S + S + S + S \\ &= 4S\end{aligned}$$



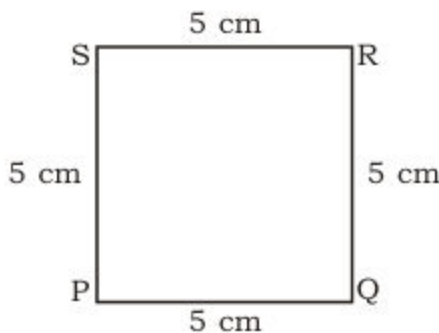
i.e. $\text{Perimeter of square} = 4 \times \text{Side}$

Where S denotes the length of a side.

Example 1. Find the perimeter of a square whose each side is 5 cm long.

Solution: Here side = 5 cm

$$\begin{aligned}\text{Now Perimeter} &= 4 \times \text{side} \\ &= 4 \times 5 \\ &= 20 \text{ cm}\end{aligned}$$



Remember that

$$\text{Side of square} = \frac{\text{Perimeter of square}}{4}$$

Example 2. Find the length of side of a square whose perimeter is 40 cm.

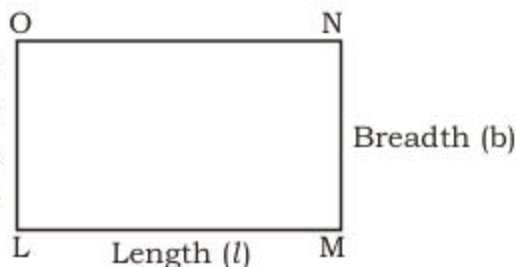
Solution: Here perimeter = 40 cm

$$\begin{aligned} \text{We know that side of square} &= \frac{\text{Perimeter of square}}{4} \\ &= \frac{40}{4} = 10 \text{ cm} \end{aligned}$$

So, length of each side is **10 cm**.

Perimeter of rectangle:

We know that the sum of the measures of all sides of a rectangle is called its perimeter. In the figure length and breadth of rectangle are L and B respectively.



Now

$$\begin{aligned} \text{Perimeter} &= \text{Length} + \text{Breadth} + \text{Length} + \text{Breadth} \\ &= l + b + l + b \\ &= 2l + 2b = 2(l + b) \end{aligned}$$

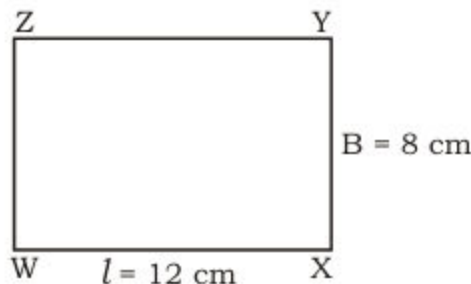
So, Perimeter of rectangle = $2(l + b)$

Example 3. Find the perimeter of rectangle whose length and breadth are 12 cm and 8 cm respectively.

Solution: Here $l = 12 \text{ cm}$

$$b = 8 \text{ cm}$$

$$\begin{aligned} \text{Perimeter of rectangle} &= 2(l + b) \\ &= 2(12 + 8) \\ &= 2(20) \\ &= 40 \text{ cm} \end{aligned}$$



So, the required perimeter of given rectangle is 40 cm.

Example 4. Find the length of rectangle whose breadth is 6 cm and its perimeter is 28 cm.

Solution: Here $l = ?$

$$b = 6 \text{ cm}$$

$$\text{Perimeter of rectangle} = 28 \text{ cm}$$

Now $\text{Perimeter of rectangle} = 2(l + b)$

$$\text{i.e. } 28 = 2(l + 6)$$

$$\text{or } \frac{28}{2} = l + 6$$

$$\text{or } 14 = l + 6$$

$$\text{or } 14 - 6 = l$$

$$\text{or } 8 \text{ cm} = l$$

$$\text{or } l = 8 \text{ cm}$$

So, the required length is **8 cm**.

Area of square:

We know that:

$$\text{Area of square} = \text{side} \times \text{side}$$

Example 5. Find the area of a square whose each side is 5 cm.

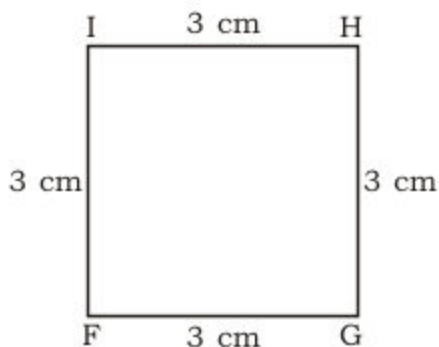
Solution: Here Side = 3 cm

We know that

$$\text{Area of square} = \text{Side} \times \text{Side}$$

$$= 3 \times 3$$

$$= 9 \text{ sq. cm}$$



Area of rectangle:

We know that:

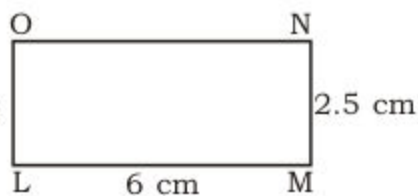
$$\text{Area of rectangle} = \text{length} \times \text{breadth}$$

Example 6. Find the area of rectangle in which length is 6 cm and breadth is 2.5 cm

Solution: Here Length = 6 cm

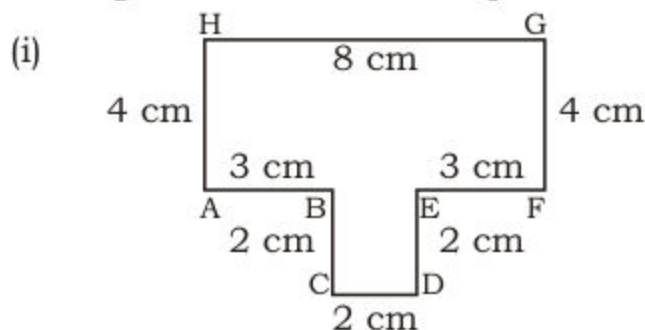
Breadth = 2.5 cm

Now Area = length $\times$ breadth = 6×2.5
 $= 15.0$ sq. cm



We can also find area and perimeter of complex figures as explained in the following example.

Example 7: Find area and perimeter of the following figures.



Solution: (i)

Perimeter of figure

According to the figure

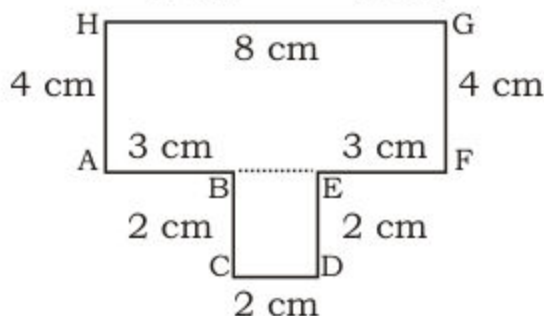
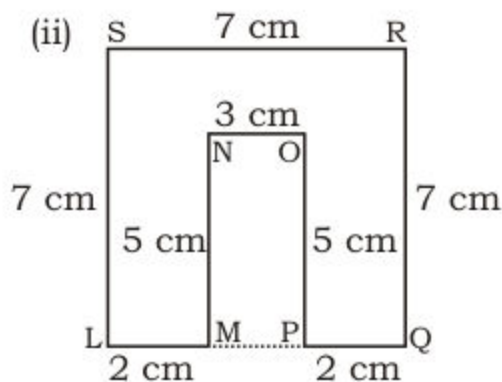
$$\begin{aligned} \text{Perimeter} &= 8 + 4 + 3 + 2 + 2 + 3 + 4 \\ &= 28 \text{ cm} \end{aligned}$$

Area of figure

In the figure there is a rectangle and square.

According to the figure:

$$\begin{aligned} \text{Area of figure} &= \text{Area of rectangle} + \text{Area of square} \\ &= 8 \times 4 + 2 \times 2 \\ &= 32 + 4 \\ &= 36 \text{ sq. cm} \end{aligned}$$



Solution: (ii)

Perimeter of figure

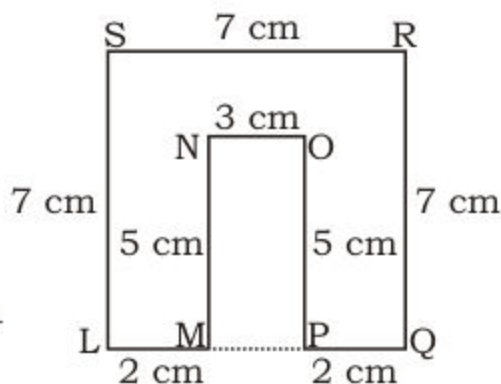
According to the figure:

$$\begin{aligned}\text{Perimeter} &= 7 + 7 + 7 + 2 + 5 \\ &\quad + 3 + 5 + 2 \\ &= 38 \text{ cm}\end{aligned}$$

Area of figure In the figure there is a square and a rectangle.

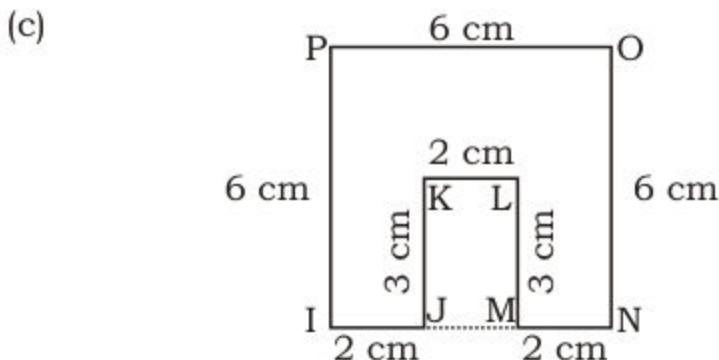
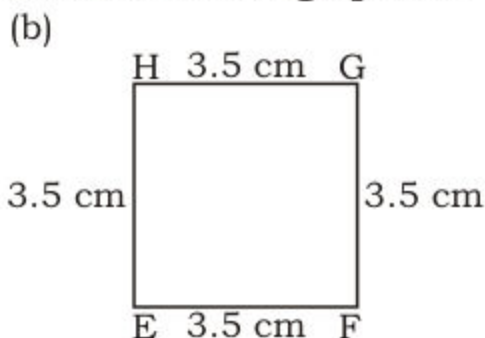
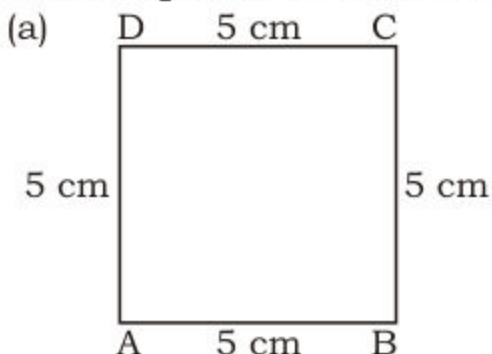
According to the figure

$$\begin{aligned}\text{Area of figure} &= \text{Area of square} - \text{Area of rectangle} \\ &= 7 \times 7 - 3 \times 5 \\ &= 49 - 15 \\ &= 34 \text{ sq. cm}\end{aligned}$$



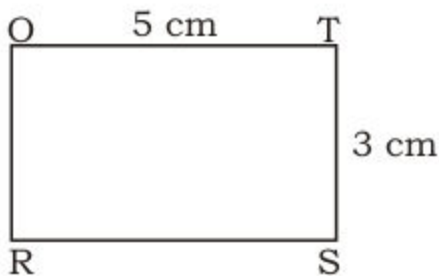
EXERCISE 11.1

1. Find the perimeter and area of each of the following squares.

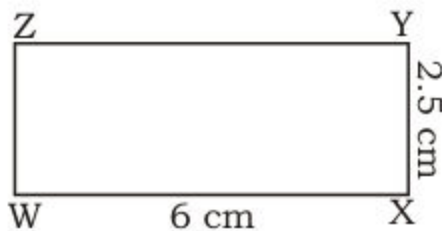


2. Find the perimeter and area of the following rectangles.

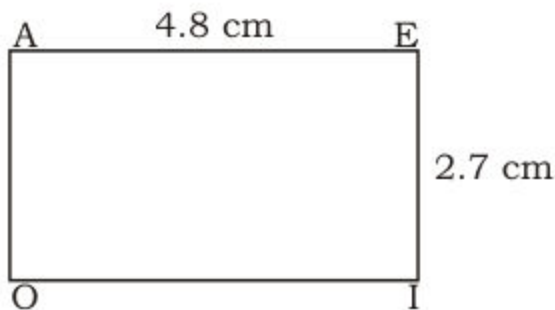
(a)



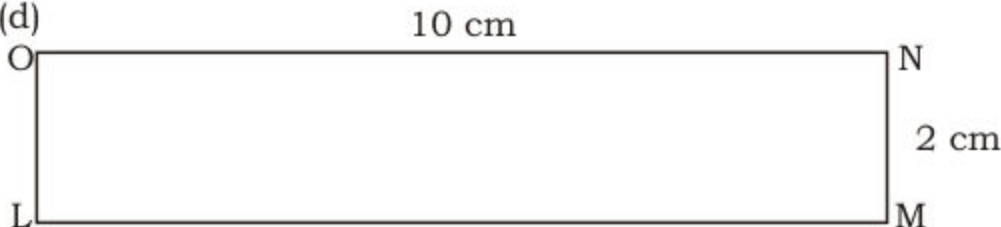
(b)



(c)



(d)



3. Find the perimeter and area of:

- (i) Rectangle with sides 5 cm, 2.5 cm, 5 cm and 2.5 cm
- (ii) Square with each side 5 cm.

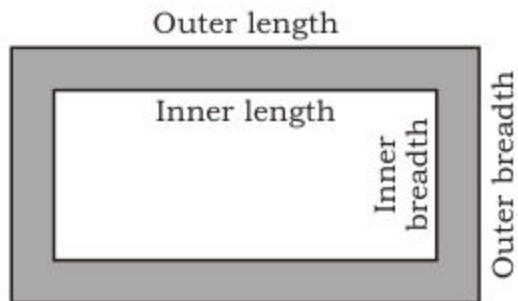
4. Find the side of square whose perimeter is:

- (a) 48 cm
- (b) 50 cm

5. Find the length of rectangle whose perimeter is 100 m and breadth is 20 m.
6. Find the area of rectangle whose adjacent sides are 6.2 cm and 8.5 cm long.

Find area of path (inside or outside) of a rectangle or square

In the adjacent figure shaded portion represents the path.



Here

Area of the large rectangle (including path)

$$= \text{Outer length} \times \text{outer breadth}$$

Also area of the small rectangle (excluding path)

$$= \text{Inner length} \times \text{Inner breadth}$$

So, $\text{Area of path} = \text{Area of large rectangle} - \text{Area of small rectangle}$

Example. A field is 60 m long and 40 m wide. A path 5 m wide is constructed all around it inside the field. Find the area of the path.

Solution:

Area of large rectangle

$$= l \times b$$

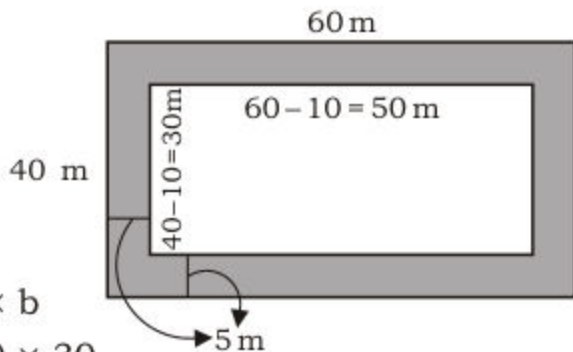
$$= 60 \times 40$$

$$= 2400 \text{ sq. m}$$

Area of small rectangle = $l \times b$

$$= 50 \times 30$$

$$= 1500 \text{ sq. m}$$



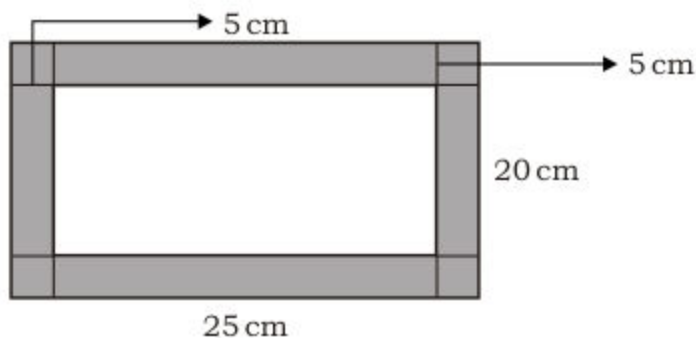
So, Area of path = $2400 - 1500$

$$= \mathbf{900 \text{ sq. m}}$$

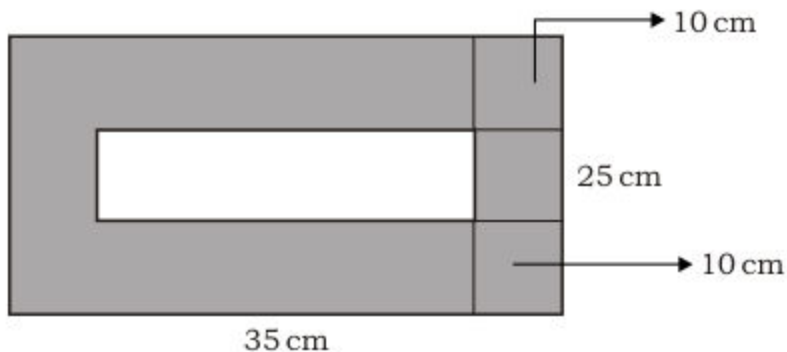
EXERCISE 11.2

1. Find the area of shaded portion in the following figures.

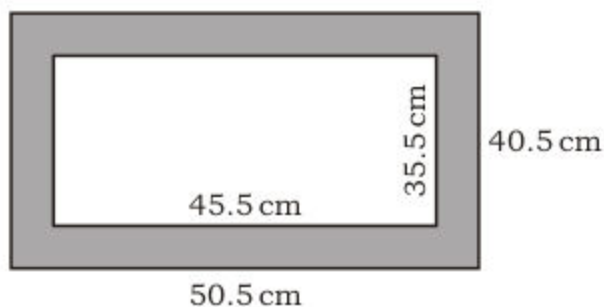
(i)



(ii)



(iii)



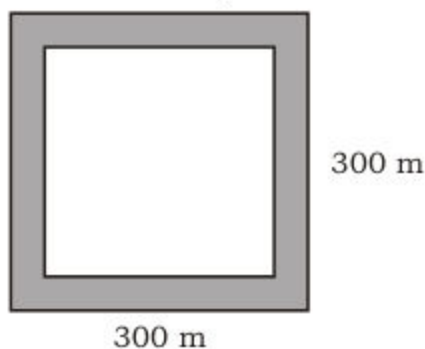
2. A field whose length and breadth are 35 cm and 30 cm respectively. A path 5 cm wide constructed all round inside it. Find the area of the path.
3. A path of 5 cm wide is constructed all round outside a field with dimension 20 cm and 10 cm. Find the area of the path.

Solve real life problems related to perimeter and area of a square and rectangle

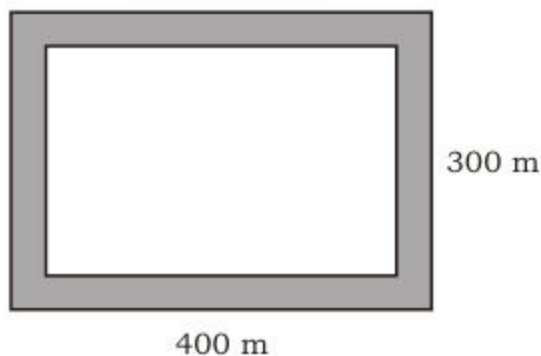
Example 1. Danish and Rafay go for jogging every morning. Danish goes around a rectangular field with length 400 m and breadth 300 m. Rafay goes around a square of 300 m length. Who covers more distance.

Solution:

Jogging track
for Rafay



Jogging track
for Danish



$$\begin{aligned}\text{Distance covered by Rafay} &= \text{Perimeter of square} \\ &= 4 \times \text{side} \\ &= 4 \times 300 \\ &= 1200 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Distance covered by Danish} &= \text{Perimeter of rectangle} \\ &= 2(l + b) \\ &= 2(400 + 300) \\ &= 2(700) \\ &= 1400 \text{ m}\end{aligned}$$

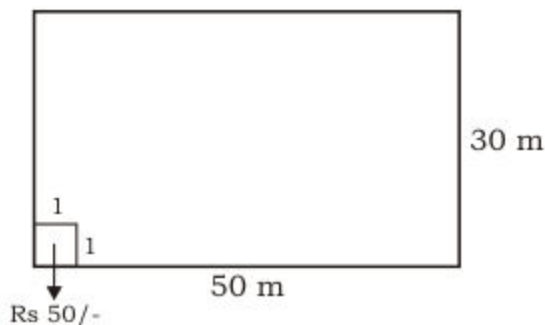
So, Danish covers more distance than Rafay.

Example 2. Find the cost of ploughing a piece of land 50 m long and 30 m wide at the rate of Rs 50 per sq. m.

Solution:

$$\begin{aligned}\text{Area of land} &= \text{Length} \times \text{breadth} \\ &= 50 \times 30 \\ &= 1500 \text{ sq. m}\end{aligned}$$

$$\begin{aligned}\text{Cost of ploughing} &= 1500 \times 50 \\ &= \textbf{75000 rupees}\end{aligned}$$



EXERCISE 11.3

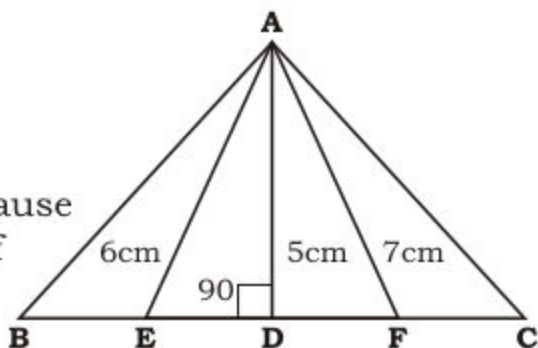
- Find the cost of fencing a square shaped park of side 250 m at the rate of Rs 30 per metre.
- Find the cost of fencing a rectangular park of length 200 m and breadth 150 m at the rate of Rs 30 per metre.
- Bismah and Umair goes for jogging every morning. Bismah covers distance around a square of length 200 m and Umair goes around a rectangular field with length and breadth of 150 m and 100 m respectively. Who covers more distance.
- Find the length of a plot of width 15 m whose area is 615 sq. m.
- The length and breadth of a play ground are 35.5 m and 30.5 m respectively. Find the cost of grassing the ground at the rate of 150 sq. m.
- A garden 120 m long and 8 m wide has a road 3 m wide all around on the garden outside. Find the cost of paving the road at the rate of Rs 200 per sq. m.
- Find the area of the floor of a room which is covered by 650 tiles, each tile has area of 0.5 sq. m.

Recognize altitude of a geometric figure as the measure of the shortest distance between the base and its top

Altitude is the shortest distance from the top to the base of a geometrical figure.

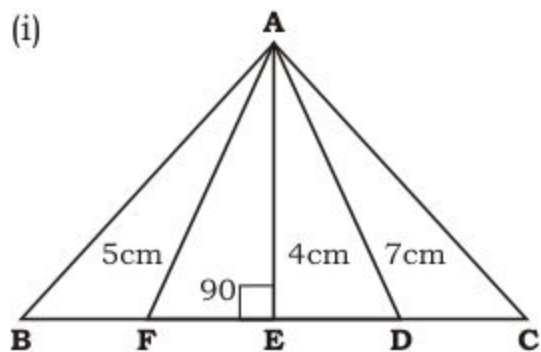
In the adjacent figure point A is the top and $\overline{BC}$ is base.

Here $\overline{AD}$ represents altitude because it shows the shortest distance of 5cm from top to the base.

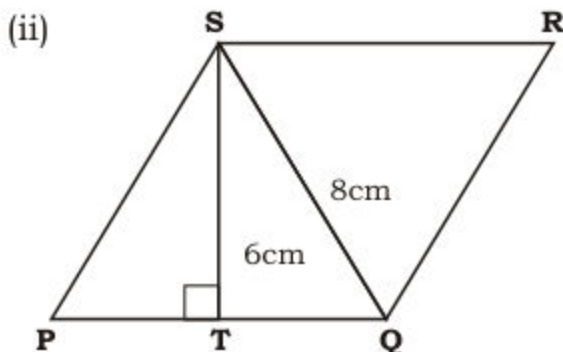


Remember that: Altitude makes right angle with the base

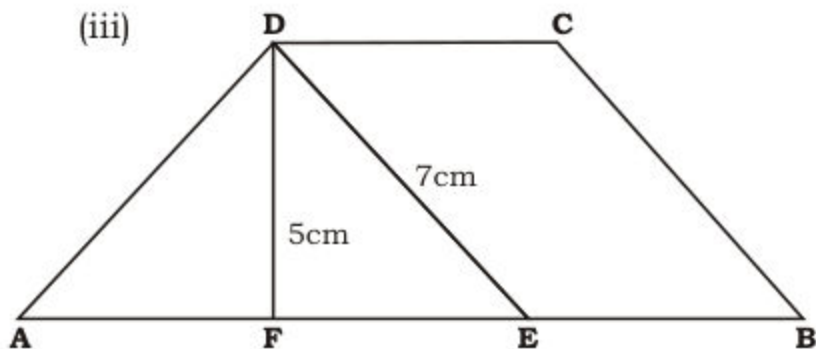
Example. Identify the altitudes in the following figures.



Altitude is $\overline{AE}$ of 4 cm



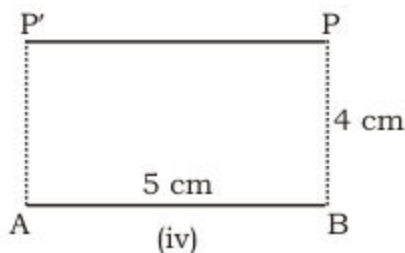
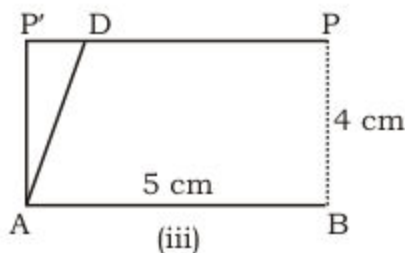
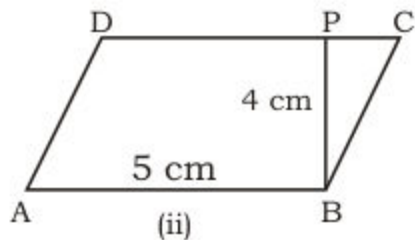
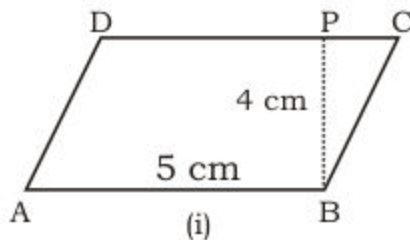
Altitude is $\overline{ST}$ of 6 cm



Altitude is $\overline{DF}$ of 5 cm

Find area of a parallelogram when altitude and base are given

Consider the following figure of a parallelogram A B C D

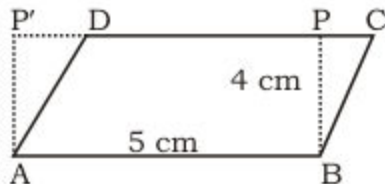


In figure (i) a perpendicular $\overline{BP}$ is drawn on the base $\overline{AB}$

In figure (ii) the triangular area PBC is being cut and separated.

In figure (iii) this triangular area is placed with $\overline{AD}$. In this way we get a rectangular area ABPP' which is shown in figure (iv)

These activities show that a parallelogramic area can be converted into rectangular area. Since we know how to find the area of rectangle, So, it is also easy to find the area of parallelogram.



$$\begin{aligned}\text{Area of rectangle} &= \text{length} \times \text{breadth} \\ &= m \overline{AB} \times m \overline{BP}\end{aligned}$$

If the value of line segment $\overline{BP}$ is known then the area of parallelogramic region can be found. The line segment BP is called the “altitude” of the parallelogram.

So, area of the parallelogram in above figure is = base $\times$ altitude
 $= 5 \times 4 = 20$ sq. cm

Here $\overline{AB}$ is called the base of the parallelogram.

Hence *area of a parallelogram = base $\times$ altitude*

If h represents altitude and b represents base then

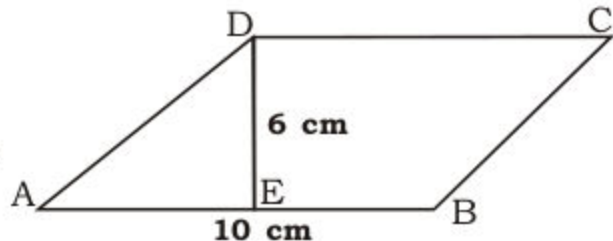
Area of parallelogram = $b \times h$

Example 1. Find the area of parallelogram whose base is 10 cm and altitude is 6 cm.

Solution: Here

$$\text{Base} = 10 \text{ cm}$$

$$\text{Altitude} = 6 \text{ cm}$$



Now,

$$\begin{aligned} \text{Area of parallelogram} &= b \times h \\ &= 10 \times 6 \\ &= 60 \text{ sq. cm} \end{aligned}$$

Example 2. The area of parallelogram is 60 sq. cm and its base is 5 cm. Find its altitude.

Solution: Here

$$\text{Base} = 5 \text{ cm}$$

$$\text{Area} = 60 \text{ sq. cm}$$

$$\text{Altitude} = ?$$

We know that

$$\begin{aligned} \text{Area of parallelogram} &= b \times h \\ \text{i.e. } 60 &= 5 \times h \end{aligned}$$

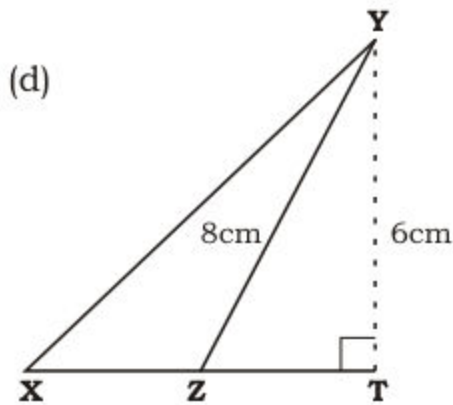
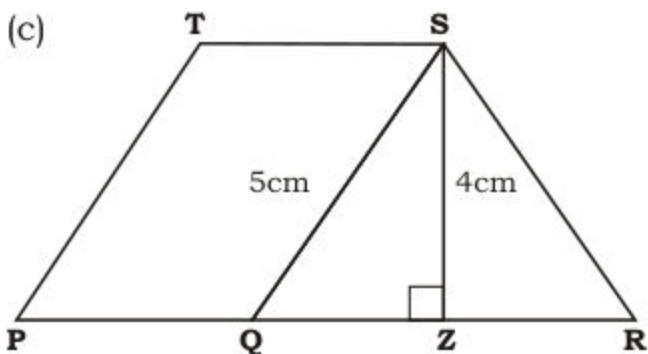
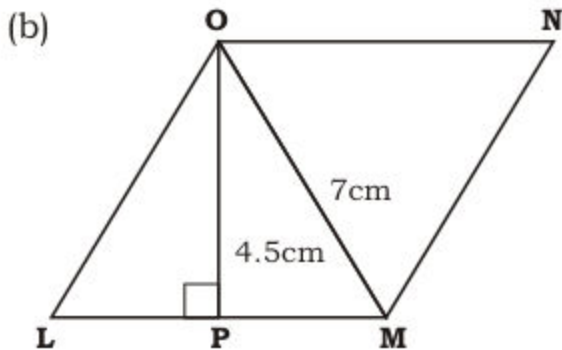
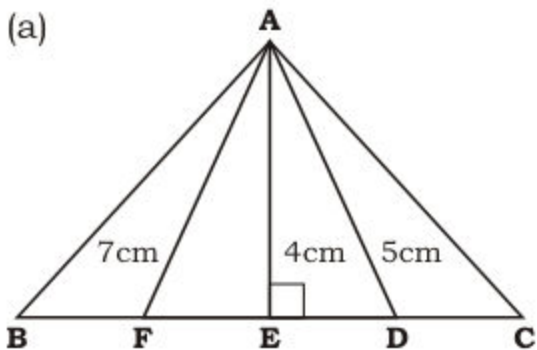
$$\frac{60}{5} = h$$

$$12 = h$$

So, its altitude is **12 cm** long.

EXERCISE 11.4

1. Identify altitude with its length in the following figures.



2. Find the area of a parallelogram when its altitude and base are given:

- Base = 10 cm, altitude = 6 cm
- Base = 15 cm, altitude = 8 cm
- Base = 4.5 cm, altitude = 5.5 cm
- Base = 6.5 cm, altitude = 4.7 cm

3. The area of a parallelogram is 192 sq. cm and its altitude is 12 cm. Find its base.

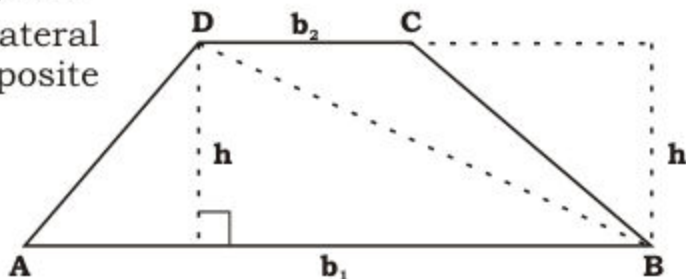
4. The base of parallelogram is 15 m and its area is 180 sq. m. Find its altitude.

5. The altitude of a parallelogram is 1.5 cm and its area is 4.50 sq. cm. Find its base.

Define trapezium and find its area when altitude and measures of the parallel sides are given

A trapezium is a quadrilateral whose only one pair of opposite sides is parallel.

The adjacent trapezium ABCD is divided into two triangles ABD and BCD.



Area of trapezium ABCD = Area of $\triangle ABD$ + Area of $\triangle BCD$

$$= \frac{1}{2}h \times b_1 + \frac{1}{2}h \times b_2$$

$$= \frac{1}{2}h (b_1 + b_2)$$

Where b_1, b_2 are the lengths of parallel sides and
 h = altitude or height

So, Area of trapezium = $\frac{1}{2} \times \text{altitude} \times (\text{sum of lengths of parallel sides})$

Example. Find the area of trapezium with parallel sides of lengths 6 cm and 10 cm and altitude is 5 cm.

Solution: Here

Altitude = 5 cm

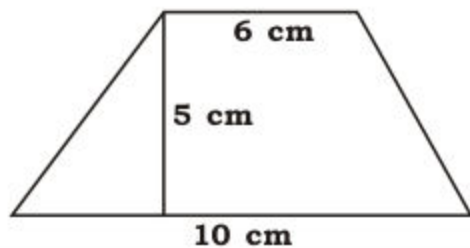
Parallel sides are 6 cm and 10 cm long.

Area of trapezium = $\frac{1}{2} \times \text{altitude} \times (\text{sum of parallel sides})$

$$= \frac{1}{2} \times 5 (6 + 10)$$

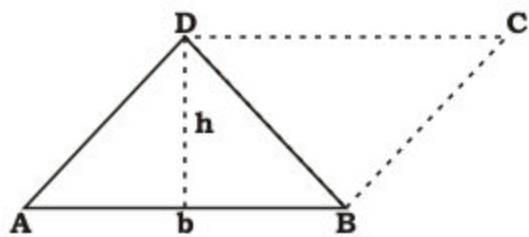
$$= \frac{5}{2} \times 16 = 5 \times 8$$

$$= 40 \text{ sq. cm}$$



Find area of a triangle when measures of the altitude and base are given.

Look at the parallelogram ABCD which is divided into two equal triangles ABD and BCD.



So,

Area of $\triangle ABD$ = Half of area of parallelogram

$$= \frac{1}{2} (\text{Base} \times \text{altitude})$$

$$= \frac{1}{2} b \times h$$

Where b = base and h = altitude or height

Example 1. Find the area of triangle in which altitude and base are 20 cm and 8 cm respectively.

Solution: Here

$$\text{Base} = 20 \text{ cm}$$

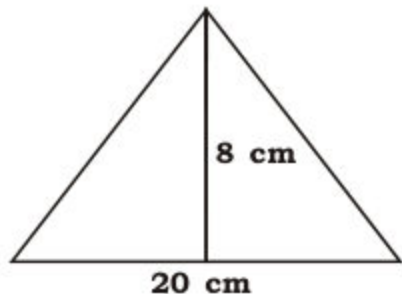
$$\text{Altitude} = 8 \text{ cm}$$

We know that

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{altitude}$$

$$= \frac{1}{2} \times 20 \times 8$$

$$= 80 \text{ sq. cm}$$



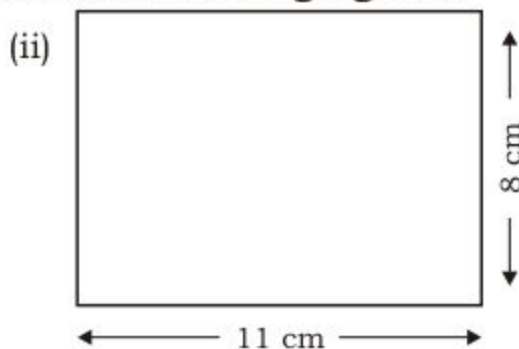
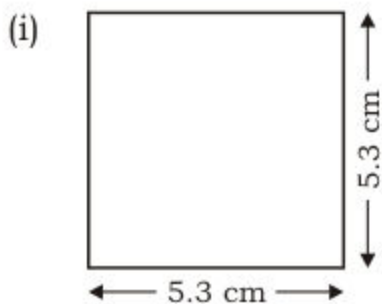
EXERCISE 11.5

1. **Find the area of trapezium when parallel sides and altitude are given as under.**
 - (i) Parallel sides are of 5 cm and 6 cm where altitude is 4 cm.
 - (ii) Parallel sides are of 3.5 cm and 4.5 cm where altitude is 2 cm.
 - (iii) Parallel sides are of 4.5 cm and 6.5 cm where altitude is 5.5 cm.
2. The parallel sides of trapezium are 6 cm and 8 cm, its altitude is 5 cm. Find its area.
3. If the parallel sides of trapezium are 15 cm and 20 cm and its area is 140 sq. cm. Find the altitude.
4. **Find the area of triangle when its base and altitude are given as under.**
 - (i) Base = 8 cm, altitude = 6 cm
 - (ii) Base = 9 cm, altitude = 5 cm
 - (iii) Base = 7 cm, altitude = 2.5 cm
 - (iv) Base = 6.5 cm, altitude = 3.5 cm
5. Find the area of triangle whose base and altitude are 6 cm and 5.5 cm respectively.
6. The area of triangle with base 5 cm is 40 sq. cm. Find its height.
7. Find the area of parallelogram in the figure if area of shaded triangle is 20 sq. m.

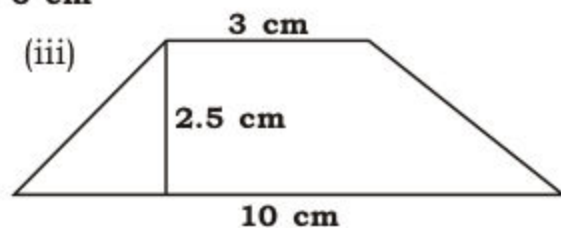
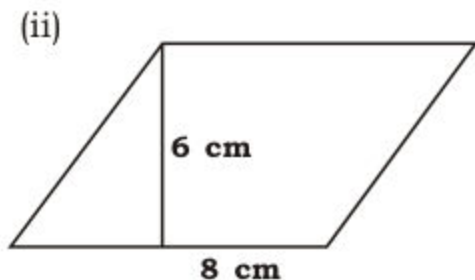
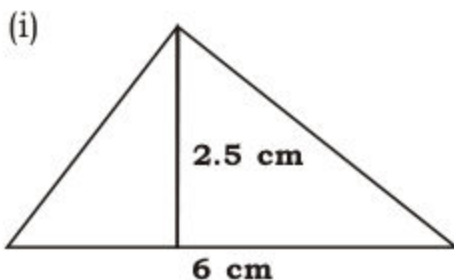


REVIEW EXERCISE 11

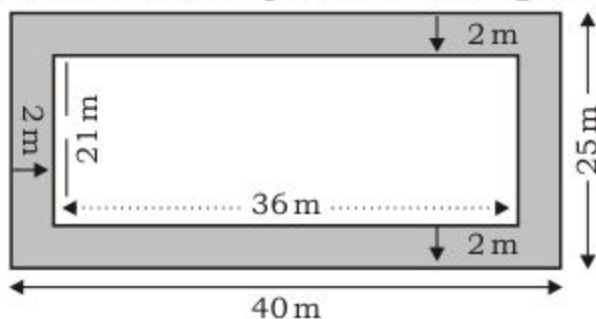
1. Find the perimeter and area of the following figures:



2. Find the area of the following figures:



3. Find area of the shaded portion of the given figure.

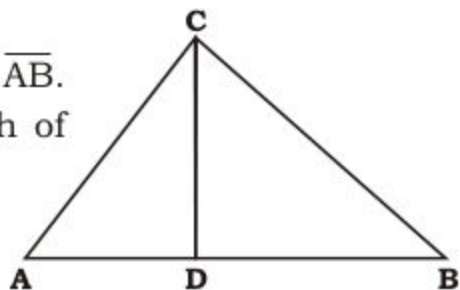


4. A tile is a square of side 20 cm. How many such tiles would be required to cover the floor of a square bath-room of side 3 m?

5. The area of a square of side 16 cm is the same as that of a rectangle of length 64 cm. What is the breadth of the rectangle?
6. The area of a triangle is 72 sq. cm. If its height is 8 cm, find its base.
7. The area of a triangle is 60 sq. cm. If its base is 12 cm, find its altitude.
8. The sides of a right angled triangle forming the right angle are 12 cm and 9 cm respectively. Find the area of that triangle?

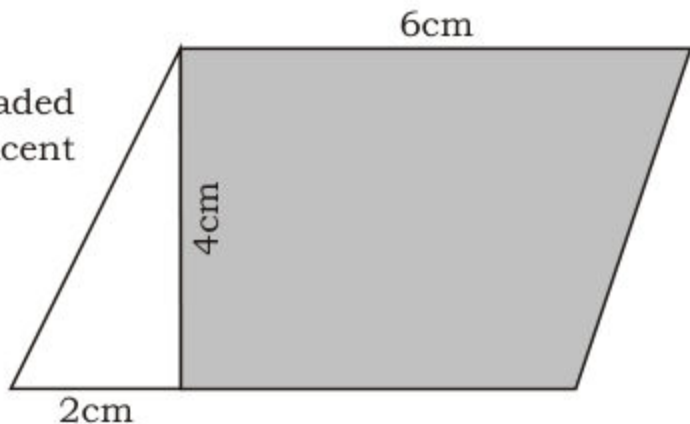
9. In $\triangle ABC$, $\overline{CD}$ is perpendicular to $\overline{AB}$. such that $m \overline{AB} = 18$ cm, length of $\overline{CD} = 7.5$ cm.

Find area of the $\triangle ABC$.



10. The parallel sides of a trapezium are 9 cm and 7 cm respectively. Its height is 5 cm. Find its area?
11. If the parallel sides of a trapezium are 20 cm and 30 cm and its area is 200 sq. cm. Find its height.
12. Area of a trapezium is 120 sq. cm. If its height is 8 cm and one of the parallel sides is 20 cm, find the length of the other side.

13. Find the area of shaded portion in the adjacent figure.



14. Fill in the blanks.

- (i) Area is commonly measured in _____.
- (ii) Perimeter is commonly measured in _____.
- (iii) Area of a rectangle is the _____ of the length and breadth.
- (iv) Perimeter of a rectangle is the _____ of its 4 sides.
- (v) Area of a square is the product of _____ and _____.
- (vi) Perimeter of a square = _____ $\times$ _____.
- (vii) Formula for finding the perimeter of a rectangle is _____.
- (viii) Formula for finding area of a parallelogram is _____.
- (ix) Formula for finding area of a triangle is _____.
- (x) Area of a trapezium is _____.

SUMMARY

- Perimeter of a Rectangle = $2 \times (L + B)$.
- Perimeter of a square = $4 \times \text{side}$
and side of a square = $\frac{\text{Perimeter}}{4}$
- Area of a Rectangle = $(L \times B)$ sq. unit
- Area of a square = $(\text{side} \times \text{side})$ sq. unit
- Area of a parallelogram = $(\text{Base} \times \text{Altitude})$ sq. unit
- Area of a Triangle = $\frac{1}{2} \times (\text{Base} \times \text{Altitude})$ sq. unit
- Area of a Trapezium = $\frac{1}{2} \times (\text{sum of parallel sides}) \times \text{altitude}$ sq. unit